



# INERTIAL SENSE

autonomous navigation solutions

## IMX-5 (IMU, AHRS, GNSS-INS)

Tactical Grade Inertial Systems

+RTK +Dual GNSS

### Overview

The **IMX-5™** is a 10-DOF sensor module consisting of a tactical grade Inertial Measurement Unit (IMU), magnetometer, and barometer. Output includes angular rate, linear acceleration, magnetic vector, and barometric pressure and altitude. IMU calibration consists of bias, scale factor, cross-axis alignment, and temperature compensation. The IMX-5 includes Attitude Heading Reference System (**AHRS**) sensor fusion to estimate roll, pitch, and heading. Adding GNSS input to the IMX-5 enables onboard Inertial Navigation System (**INS**) sensor fusion for roll, pitch, heading, velocity, and position.

The **RUG-3-IMX-5™** series adds a rugged aluminum enclosure and RS232, RS485, and CAN bus to the IMX-5.

The **RUG-3-IMX-5-RTK™** includes a multi-frequency GNSS receiver with RTK precision position enabling INS sensor fusion for roll, pitch, heading, velocity, and position.

The **RUG-3-IMX-5-Dual™** includes two multi-frequency GNSS receivers with RTK precision position and dual GNSS heading/compass.

The **Inertial Sense SDK** is an open-source software development kit for quick integration to configure and communicate with Inertial Sense products. The SDK includes data logger, math libraries, and interface for Linux, Windows, and embedded platforms.



**RUG-3-IMX-5**

Size: 30.5 x 25.4 x 9.9 mm

Weight: 10.5 g



**IMX-5**

Size: 15.6 x 12.5 x 2.9 mm

Weight: 0.8 g

INS: External GNSS Input



**RUG-3-IMX-5-RTK/Dual**

Size: 30.5 x 25.4 x 14.8 mm

Weight: 14 g

GNSS: Multi-Band L1/L2/E5

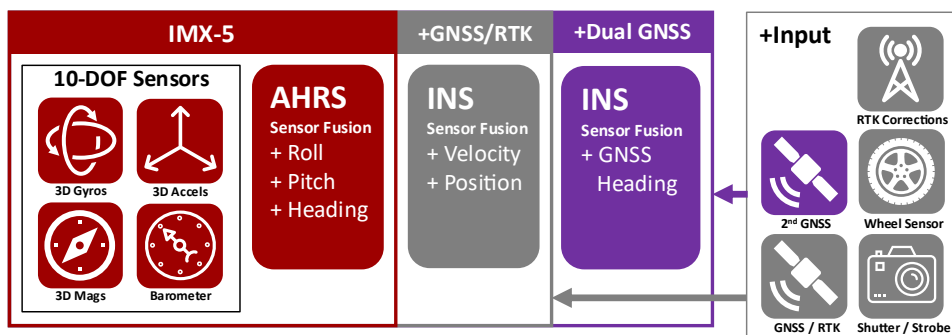
### Features

- **Tactical Grade IMU**
  - Gyro: 2.0 °/hr Bias Instability, 0.23 °/vhr ARW
  - Accel: 27 µg Bias Instability, 0.020 m/s/vhr VRW
- **0.04° Dynamic Roll/Pitch**
- **0.13° Dynamic Heading**
- **Surface Mount Reflowable (PCB Module)**
- **Output Data Rates:**
  - 1000Hz IMU, 200Hz AHRS, 142Hz GNSS-INS
- External GNSS Input Support (Multi-Band)
- Attitude (Roll, Pitch, Yaw, Quaternions), Velocity, and Position UTC Time Synchronized
- Triple Redundant IMUs Calibrated for Bias, Scale Factor, Cross-axis Alignment, and G-sensitivity
- -40°C to 85°C Sensor Temperature Calibration
- Binary and NMEA ASCII Protocol
- Barometric Pressure and Humidity
- Strobe In/Out Data Sync (Camera Shutter Event)
- Fast Integration with SDK and Example Software
- Data Logging (SDK and Application Software)
- RUG-3-IMX-5: RS232, RS485, CAN\* bus

\* Available in future firmware release.

### Applications

- Drone Navigation
- Unmanned Vehicle Payloads
- Ground and Aerial Survey
- Automotive Navigation
- Stabilized Platforms
- Antenna and Camera Pointing
- First Responder and Trackers
- Robotics and Ground Vehicles
- Maritime





## Specifications

| Performance (AHRS, INS, RUG-3)       | Typ   |                                     |
|--------------------------------------|-------|-------------------------------------|
| INS Dynamic Roll/Pitch** (RMS)       | 0.04° | (GNSS-aided with motion excitation) |
| Static Roll/Pitch (RMS)              | 0.1°  |                                     |
| INS Dynamic Heading** (RMS)          | 0.13° | (GNSS velocity-derived)             |
| Static Heading w/Dual Compass* (RMS) | 0.4°  |                                     |
| Static Heading w/magnetometer (RMS)  | 1.0°  |                                     |

\*1 m baseline distance between GNSS antennas.

\*\*With GNSS input and periodic motion >0.8 m/s<sup>2</sup> acceleration and >2 m/s velocity.

| Performance (INS, RUG-3)      | RUG-3     | +RTK             |
|-------------------------------|-----------|------------------|
| Horizontal Position (w/ SBAS) | 1.5 m CEP | 1 cm + 1 PPM CEP |
| Velocity (GPS and INS)        | 0.03 m/s  |                  |
| Angular Resolution            | 0.05°     |                  |
| Operation Limits              |           |                  |
| Velocity (external GNSS)      | 500 m/s   |                  |
| Altitude (external GNSS)      | 50 Km     |                  |
| Altitude (Barometric)         | 10 Km     |                  |
| GNSS cold start time to fix   | 24 s      | -                |

| Performance                                | Typ               |
|--------------------------------------------|-------------------|
| Startup Time                               | 0.8 s             |
| INS/AHRS Timestamp Accuracy (RMS)          | 1 us              |
| Max Output Data Rate (IMU, AHRS, GNSS-INS) | 1000, 200, 142 Hz |
| IMU signal latency @ 1KHz ODR              | 4 ms              |

| Absolute Maximum Ratings        | MAX                        |
|---------------------------------|----------------------------|
| Acceleration                    | 10,000 g                   |
| Ambient Operating Temperature   | -40 to 85 °C               |
| Junction Temperature            | -40 to 105 °C              |
| Storage Temperature             | -40 to 125 °C              |
| Overpressure                    | 600 kPa                    |
| ESD rating (Human body model)   | ± 2 kV                     |
| Solder Reflow Temperature Max   | 245 °C                     |
| Solder Reflow Temperature Limit | 217 °C liquidus: 40 – 60 s |

| Sensors                     | IMU - Gyros    | IMU - Accels  | Mags      | Pressure    |
|-----------------------------|----------------|---------------|-----------|-------------|
| Operating Range             | ±4000 °/sec    | ±16 g         | ±2500 μT  | 30–125 kPa  |
| In-Run Bias Stability*      | < 2.0 °/hr     | < 27 μg       |           |             |
| Random Walk: ARW, VRW       | 0.23 °/√hr     | 0.020 m/s/√hr |           |             |
| Non-linearity               | 0.02 % FSR     | 0.02 % FSR    |           |             |
| Noise Density               | 5 mdps/√Hz     | 60 μg/√Hz     | 20 nT/√Hz | 0.18 Pa/√Hz |
| Bias Error over -40C to 85C | 0.3 °/s RMS    | 3.7 mg RMS    |           |             |
| Max Output Rate             | 1 KHz          | 1 KHz         | 50 Hz     | 50 Hz       |
| Bandwidth                   | 539 Hz         | 416 Hz        | 50 Hz     | 5 Hz        |
| Alignment Error             | 0.03°          | 0.03°         | 0.05°     |             |
| Resonant Frequency          | 20 KHz         | 2.6/2.17 KHz  |           |             |
| Sampling Rate               | 8 KHz          | 4 KHz         | 300 Hz    | 200 Hz      |
| Resolution                  | **0.0076 °/sec | **122 μg      | 0.3 μT    | 0.03 Pa     |

\*Performance specifications per IEEE-STD-952-1997

\*\*1 KHz resolution after oversampling

| Function                          | IMX™ | +RTK | +Dual |
|-----------------------------------|------|------|-------|
| Gyro & Accelerometer (IMU)        | •    | •    | •     |
| Magnetometer & Barometer          | •    | •    | •     |
| Roll, Pitch, Heading (AHRS)       | •    | •    | •     |
| Heading, Velocity, Position (INS) |      | •    | •     |
| GNSS Heading                      |      |      | •     |



Development  
Kits available on  
our website.

## Electrical (IMX-5)

| Power Draw                     | Min  | Typ | Max | Units |
|--------------------------------|------|-----|-----|-------|
| IMU @ 1KHz                     |      | 95  | 105 | mW    |
| w/ AHRS, INS @ 250Hz           |      | 100 | 110 | mW    |
| Supply Voltage (Vcc)           | 3.0  | 3.3 | 3.6 | V     |
| I/O Pin MAX Voltage Range      | -0.5 |     | 3.6 | V     |
| Total Output Current, All Pins |      |     | 100 | mA    |
| I/O Pin Output Current         |      |     | 20  | mA    |
| I/O Pin Input low-level        | 0.99 |     |     | V     |
| I/O Pin Input high-level       | 2.31 | 3.3 | 3.6 | V     |
| I/O Pin Output high-level      |      | 3.3 |     | V     |
| STROBE input frequency         |      |     | 1   | KHz   |
| STROBE output jitter           |      | 10  |     | us    |
| Rising Slope of VIN*           | 2.4  |     |     | V/ms  |

\*The supply rising slope must be higher than minimum rating for proper function.

## Electrical (RUG-3)

|                           | Min | Typ  | Max | Units |
|---------------------------|-----|------|-----|-------|
| Supply Voltage (VIN)      | 4.5 |      | 20  | V     |
| RUG-3-IMX-5-RTK + Antenna |     |      |     |       |
| Current Draw @ 5V, 250Hz* |     | 185  |     | mA    |
| Power Consumption @250Hz* |     | 927  |     | mW    |
| Power Consumption @100Hz* |     |      |     | mW    |
| Power Consumption – Dual  |     | 1470 |     | mW    |

\*Navigation filter update rate.

## Mechanical (IMX-5)

|                                     | Typ               | Units |
|-------------------------------------|-------------------|-------|
| Size                                | 15.6 x 12.5 x 2.9 | mm    |
| Weight                              | 0.8               | grams |
| Effective Thermal Resistance (Θeff) | 19                | °C/W  |
| Thermal Resistance (ΘJA, ΘJB, ΘJC)  | 53, 34.5, 2.0     | °C/W  |

## Mechanical (RUG-3)

|              |                                              | Units | Conditions              |
|--------------|----------------------------------------------|-------|-------------------------|
| Size         | 30.5 x 25.4 x 9.9<br>30.5 x 25.4 x 14.8      | mm    | RUG-3<br>RUG-3-RTK/Dual |
| IP Rating    | 40                                           |       | No liquid protection    |
| Mounting Tab | 30.836                                       | mm    |                         |
| Hole Spacing |                                              |       |                         |
| Weight       | 14.0                                         | grams |                         |
| Connectors   | Main: Harwin# G125-MV11205L1P, GPS 1/2: MMCX |       |                         |

## Communications & I/O

|                         |                                       |
|-------------------------|---------------------------------------|
| IMX-5 Interface         | USB, UART x3, SPI                     |
| RUG-3 Interface         | USB, UART x2, RS232, RS485, CAN*, SPI |
| Max Baud Rate:          |                                       |
| SPI                     | 10 Mbps                               |
| UART, RS422, RS485      | 10 Mbps                               |
| RS232                   | 500 Kbps                              |
| Strobe Inputs / Outputs | 4 / 1                                 |

\* Available in future firmware release.

